Welcome to celestial holography

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July 2026

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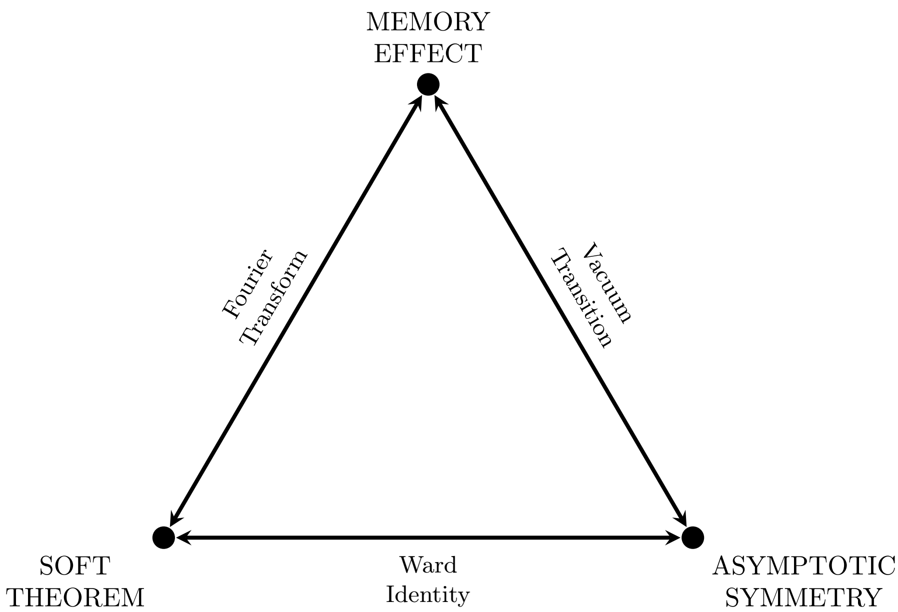
1 Lecture 1. Introduction: Infrared triangle and CFT

Celestial Holography, as follows from the name, uses the idea of holographic duality, which means establishing correspondence between symmetries in the bulk of the space-time (in our case symmetries of asymptotically flat space-time) and symmetries on its boundary (in our case 2-dimensional conformal symmetry). Therefore, to build a bridge from bulk to boundary, we need to get familiar with each of the two subjects in particular. In the first lecture we will introduce the core developments, tools and methods used to study both asymptotically flat space-times and conformal field theory.

1.1 Infrared triangle

Creation of the celestial holography program would be unimaginable without a deep study of infrared issues arising in any QFT containing massless particles that can have low energies. Before proceeding to understand the celestial basis, we should quickly talk about this IR story, and the central figure in such discussion will be the infrared triangle.



Infrared triangle is a relation that connects 3 seemingly unrelated subjects: soft theorem, asymptotic symmetries and memory effect.

Even though these three subjects seem unrelated, any two of them are deeply connected, and in reality are the same statement expressed in different "languages".

The first vertex, soft theorems, originated in QED and was generalized to gravity in 1965 by Weinberg [1]. This family of theorems describes how scattering amplitudes factorize when a massless external particle becomes soft.

The second vertex, asymptotic symmetries, is connected to symmetries that a system has on its asymptotic boundary. Back in 1962 Bondi, van der Burg, Metzner and Sachs [2, 3] expected that asymptotically flat spacetime of general relativity would have the Poincare group of special relativity. Surprisingly, they found that the symmetry group is in fact infinite dimensional.

The third vertex, memory effect, was discovered in 1974 by Zelьdovich and Polnarev [4] and developed by Christodoulou. This effect describes how a passage of gravitational wave through a LIGO-like detector produces a permanent shift in the positions of its mirrors. Though not tested yet, there are several proposals of how to test it in the next several decades.

As depicted in the picture below, we can transition between any two of these vertices. Ward identities written down for asymptotical symmetry of the S-matrix lead to the soft theorems. If we do a Fourier transform of a soft theorem, the infrared pole will become a step function in time, leading to the memory effect. And finally, transition between two vacua, related by asymptotic symmetry, is nothing else but a memory effect.

The IR physics in 4D is not the purpose of our lectures, so we will not go into too much detail, leaving just the necessary parts that will be used in the celestial story later on. For the most expanded discussion on this matter an interested reader can refer to Strominger's lectures. However, some details and explanations I will provide in this very first lecture, concentraiting on the conceptual significance of what will be done.

1.1.1 Asymptotic symmetries

The BMS group was studied back at 1960s [2,3] by H. Bondi, M. G. J. van der Burg, A. W. K. Metzner and Rainer K. Sachs to look at the behaviour of gravitational waves at null infinity, finding an infinite dimensional set of supertranslations belonging to the symmetry group. Half a century has passed and the interest to description of asymptotically flat spacetimes began to grow and it was shown in [5,6] how supertranslation symmetry implies corresponding Ward identity and extensions of BMS group have been proposed [7,8].

To begin with, some short reminder of asymptotically flat spacetimes and their symmetries will be brought up.

Definition 1. *We will call a spacetime **asymptotically flat** if we can write its metric tensor as a sum* $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$*, where* $\eta_{\mu\nu}$ *is a Minkowski metric and* $h_{\mu\nu}$ *satisfies the following conditions:*

1. $\lim_{r\to\infty} h_{\mu\nu} = O(1/r)$;
2. $\lim_{r\to\infty} \partial_\rho h_{\mu\nu} = O(1/r^2)$;
3. $\lim_{r\to\infty} \partial_\lambda \partial_\rho h_{\mu\nu} = O(1/r^3)$,

where $r:\ r^2 = x^2 + y^2 + z^2$ *is the distance from the center of coordinates.*

Basically, these conditions mean that the curvature of spacetime vanishes at large distances from the center of coordinate system and that at large distances spacetime becomes indistinguishable from Minkowski spacetime.

One might remember how we describe the asymptotics of Minkowski spacetime when coordinates go to infinity, using Penrose diagrams. We might recall it, making a coordinate transformation:

$$\begin{aligned} t + r &= \tan \frac{1}{2}(T + R), \\ t - r &= \tan \frac{1}{2}(T - R). \end{aligned} \tag{1}$$

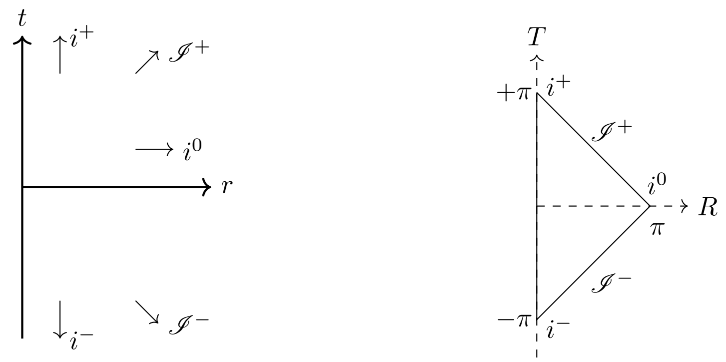


Figure 1: Minkowski space-time in spherical coordinates (t, r, θ, ϕ) and Minkowski space-time in conformally transformed spherical coordinates (T, R, θ, ϕ), where t- and r-infinities are transferred to finite distances. Here $t, r \in (-\infty, +\infty)$, and $T \in [-\pi, +\pi], R \in [0, \pi]$.

Looking at the Figure 1 we find different types on infinities on it:

1. i^+ – *timelike future infinity* with $t \to +\infty, r = const.$ Massive particles who travel with speed less than the speed of light, end up at i^+.
2. i^- – *timelike past infinity* $t \to -\infty, r = const.$ Massive particles who travel with speed less than the speed of light, originate from i^-.
3. i^0 – *spacelike infinity* $t = const, r \to \infty$. This is the area in which space-like slicing can be realised.
4. $\mathscr{I}^+$ – *null future infinity* $t + r \to \infty, t - r = const.$ Massless particles who travel with the speed of light, end up at $\mathscr{I}^+$.
5. $\mathscr{I}^-$ – *null past infinity* $t - r \to -\infty, t + r = const.$ Massless particles who travel with the speed of light, originate from $\mathscr{I}^+$.

In the following the future and past null infinities $\mathscr{I}^{\pm}$ will be the most important subjects for us of the ones listed above. Far away from the beginning of coordinates null infinities of an asymptotically flat spacetime (from where massless particles originate and where they end up) have the same structure as Minkowski spacetime. Loosely speaking, slices of these null infinities will be the boundaries where our 2D conformal field theory will dwell.

Asymptotically flat spacetime can be described in terms of a BMS formalism, starting with imposing a Bondi gauge on the form of a metric.

Definition 2. *We say that we have imposed the **Bondi gauge** if the metric meets the following conditions:*

(i) We can write metric in form

$$e^{2\beta}\frac{V}{r}du^2 - 2e^{2\beta}dudr + 2g_{uA}dudx^A + g_{AB}dx^Adx^B,$$

where

$$\begin{aligned} \frac{1}{r}V(u,x^A) &= -1 + O(1/r); \\ \beta(u,x^A) &= O(1/r^2); \\ g_{uA}(u,r,x^A) &= O(1); \\ g_{AB}(u,r,x^A) &= r^2\gamma_{AB} + O(r), \end{aligned}$$

and $\gamma_{AB}:\ ds^2 = \gamma_{AB}\, dzd\bar{z} = 4\, dzd\bar{z}/(1+z\bar{z})^2$ is a unit metric on a sphere.

(ii) The conditions are satisfied:

$$g_{rr} = g_{rA} = 0 = g^{uu} = g^{uA}, \qquad \partial_r \det\big(g_{AB}/r^2\big) = 0$$

(iii) Near $\mathscr{I}^+$ we also require fall-off conditions for stress-energy tensor:

$$\begin{aligned} T^M_{uu} &= O(r^{-2}), \quad T^M_{uA} = O(r^{-2}), \\ T^M_{ur} &= O(r^{-4}), \quad T^M_{rA} = O(r^{-3}), \\ T^M_{rr} &= O(r^{-4}), \quad T^M_{AB} = O(r^{-1}). \end{aligned}$$

The flat space coordinates and Bondi coordinates are connected via

$$X^\mu = \left(u + r, r\frac{z+\bar{z}}{1+z\bar{z}}, ir\frac{z-\bar{z}}{1+z\bar{z}}, r\frac{1-z\bar{z}}{1+z\bar{z}}\right). \tag{2}$$

The stereographic coordinates $z, \bar{z}$ are related to spherical coordinates θ, φ as follows:

$$\begin{cases} z = e^{i\varphi}\cot\dfrac{\theta}{2}, \\ \bar{z} = e^{-i\varphi}\cot\dfrac{\theta}{2}. \end{cases} \tag{3}$$

In the same manner, we impose the Bondi gauge on the past null infinity $\mathscr{I}^-$. Analogously to the ***retarded*** coordinate $u = t - r$ on $\mathscr{I}^+$, we introduce the ***advanced*** coordinate $v = t + r$.

A metric in the Bondi gauge can be reduced to the following form:

- Near $\mathscr{I}^+$

$$\begin{aligned} ds^2 = -du^2 - 2dudr + 2r^2\gamma_{z\bar{z}}dzd\bar{z} + \frac{2m_B}{r}du^2 + rC_{zz}dz^2 + rC_{\bar{z}\bar{z}}d\bar{z}^2 \\ -2U_zdudz - 2U_{\bar{z}}dud\bar{z} + \cdots, \end{aligned} \tag{4}$$

  with $U_z = -\frac{1}{2}D^zC_{zz}$;

- Near $\mathscr{I}^-$

$$\begin{aligned} ds^2 = -dv^2 + 2dvdr + 2r^2\gamma_{z\bar{z}}dzd\bar{z} + \frac{2m_B}{r}dv^2 + rD_{zz}dz^2 + rD_{\bar{z}\bar{z}}d\bar{z}^2 \\ -2V_zdvdz - 2V_{\bar{z}}dvd\bar{z} + \cdots, \end{aligned} \tag{5}$$

  with $V_z = \frac{1}{2}D^zD_{zz}$.

Here m_B is a Bondi mass aspect (since in some spacetimes it can be connected to the mass), C_{zz} and D_{zz} are defined as outgoing and incoming shear tensors, while $N_{zz} \equiv \partial_u C_{zz}$, $M_{zz} \equiv \partial_v D_{zz}$ – outgoing and incoming Bondi news tensors. All these functions may depend on $(u, z, \bar{z})$, but not on r.

Exercise 1. *Show that in these coordinates, the Minkowski metric takes the form:*

$$ds^2 = -du^2 - 2dudr + 2r^2\gamma_{z\bar{z}}dzd\bar{z}. \tag{6}$$

Actually, m_B, C_{zz}, N_z are not all independent. They are related by constraint equations. The uu constraint gives [9]

$$\partial_u m_B = \frac{1}{4}D_z^2 N^{zz} + \frac{1}{4}D_{\bar{z}}^2 N^{\bar{z}\bar{z}} - \frac{1}{2}T_{uu}, \tag{7}$$

while the uz constraint reduces to

$$\begin{aligned} \partial_u N_z = \frac{1}{4}D_z\left(D_z^2 C^{zz} - D_{\bar{z}}^2 C^{\bar{z}\bar{z}}\right) - T_{uz}^M + \partial_z m_B + \frac{1}{16}D_z\partial_u(C_{zz}C^{zz}) \\ - \frac{1}{4}\left(N^{zz}D_z C_{zz} + N_{zz}D_z C^{zz}\right) - \frac{1}{4}D_z\left(C^{zz}N_{zz} - N^{zz}C_{zz}\right), \end{aligned} \tag{8}$$

where we defined

$$T_{uu} = \frac{1}{4}N_{zz}N^{zz} + \lim_{r\to\infty} r^2 T_{uu}^M. \tag{9}$$

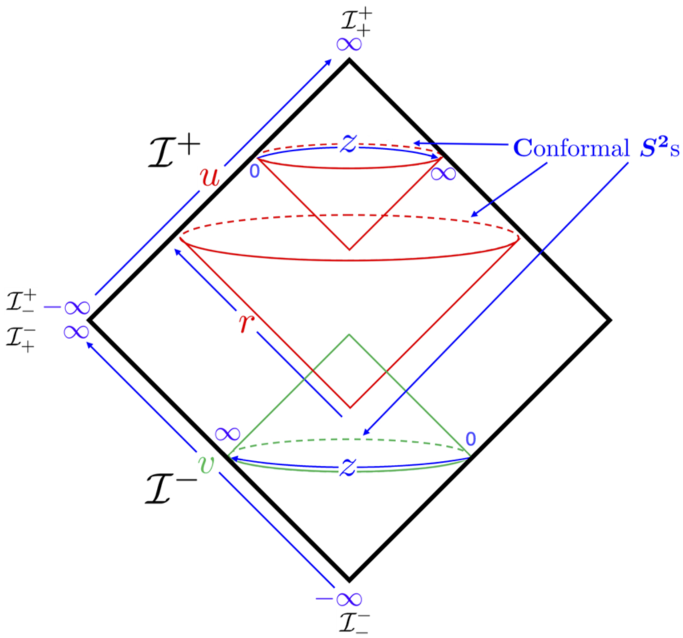


Figure 2: The structure of our asymptotically flat spacetime. $\mathscr{I}^\pm$ is denoting future and past null infinities; u and v are retarded and advanced Bondi time coordinates.

We will not perform the calculations leading to the symmetry group of (4) and its copy near $\mathscr{I}^-$ (namely, to the BMS$^\pm$ groups), as they are quite volumous. Instead, we will write down the generating vector fields, omitting the computational part.

Supertranslations near $\mathscr{I}^+$ are generated by vector fields

$$\xi^+ = f\partial_u - \frac{1}{r}(D^{\bar{z}}f\partial_z + D^z f\partial_{\bar{z}}) + D^z D_z f\partial_r, \tag{10}$$

and near $\mathscr{I}^-$ by

$$\xi^- = f^-\partial_v + \frac{1}{r}(D^{\bar{z}}f^-\partial_z + D^z f^-\partial_{\bar{z}}) - D^z D_z f^-\partial_r, \tag{11}$$

where $f(z, \bar{z}), f^-(z, \bar{z})$ – arbitary function on S^2, therefore one can clearly see that there are infinitely many supertranslations. If we take $f(z, \bar{z}) = const$, it will generate u-translations, meanwhile if we take it to be $l = 1$

harmonic on the sphere, we retrieve three spacial translations. Namely:

$$\text{x-translations :} \qquad f_x(z,\bar{z}) = \sin\theta\cos\phi = \frac{z+\bar{z}}{1+z\bar{z}}, \tag{12}$$

$$\text{y-translations :} \qquad f_y(z,\bar{z}) = \sin\theta\sin\phi = -i\frac{z-\bar{z}}{1+z\bar{z}}, \tag{13}$$

$$\text{z-translations :} \qquad f_z(z,\bar{z}) = \cos\theta = \frac{1-z\bar{z}}{1+z\bar{z}}. \tag{14}$$

Exercise 2. *Show that they indeed generate 3 spatial translations.*

To calculate, for example, how metric tensor changes under these transformations, we will need to compute its Lie derivative:

$$\delta g_{\mu\nu} = \mathcal{L}_\xi g_{\mu\nu} = \xi^\rho\partial_\rho g_{\mu\nu} + (\partial_\mu\xi^\rho)g_{\rho\nu} + (\partial_\nu\xi^\rho)g_{\mu\rho} \tag{15}$$

Therefore, the original BMS group includes Lorentz transformations and supertranslations. Later it was found that rotations can also be extended to *superrotations*, what lead to the *extended BMS group*, but this is a whole different story that we will discover during the 3rd lecture.

Since we are interested in scattering processes between $\mathscr{I}^\pm$, which are not connected together by definition, we need a prescription of initial values on them of what we are integrating. In our case we integrate m_B, therefore we need to determine the initial value. For now we propose that BMS$^+$ frame should be determined by the Lorentz- and CPT-invariant matching conditions, defining $C_{zz}|_{\mathscr{I}^+_-} = 2D_z^2 C|_{\mathscr{I}^+_-}$

$$C(z,\bar{z})|_{\mathscr{I}^+_-} = C(z,\bar{z})|_{\mathscr{I}^-_+}\,, \quad m_B(z,\bar{z})|_{\mathscr{I}^+_-} = m_B(z,\bar{z})|_{\mathscr{I}^-_+}\,. \tag{16}$$

These conditions basically "glue together" our asymptotic data on $\mathscr{I}^-_+$ and $\mathscr{I}^+_-$. Instead of having two independent symmetry groups $BMS^\pm$, we connect them through this condition and have one.

The matching condition for N_z is discussed below. As mentioned, conditions (16) break the combined BMS$^+\times$BMS$^-$ action on $\mathscr{I}^+$ and $\mathscr{I}^-$ down to the diagonal subgroup that preserves these conditions, namely,

$$f(z,\bar{z})|_{\mathscr{I}^+_-} = f(z,\bar{z})|_{\mathscr{I}^-_+}\,. \tag{17}$$

These infinite number of matching conditions yield the existence of infinite number of conserved charges. The supertranslation charges at $\mathscr{I}^+$ can be written as [10, 11]

$$Q_f^+ = \frac{1}{4\pi G}\int_{\mathscr{I}^+_-} d^2z\gamma_{z\bar{z}} f m_B, \tag{18}$$

$$Q_f^- = \frac{1}{4\pi G}\int_{\mathscr{I}^-_+} d^2z\gamma_{z\bar{z}} f m_B. \tag{19}$$

Integrating by parts, using the constraint equation, and assuming m_B decays to zero in the far future, we can write

$$Q_f^+ = \frac{1}{16\pi G}\int dud^2z f\gamma_{z\bar{z}}N_{zz}N^{zz} - \frac{1}{8\pi G}\int d^2z\gamma^{z\bar{z}} fD_z^2D_{\bar{z}}^2N, \tag{20}$$

$$Q_f^- = \frac{1}{16\pi G}\int dvd^2z f^-\gamma_{z\bar{z}}M_{zz}M^{zz} + \frac{1}{8\pi G}\int d^2z\gamma^{z\bar{z}} f^- D_z^2D_{\bar{z}}^2. \tag{21}$$

Of course the charges $Q^\pm$ existed even before imposing matching conditions. But the fact that a conservation law $Q_f^+ = Q_f^-$ takes place only if the matching conditions are set.

1.1.2 Soft theorem

The Quantum Field Theory description of scattering processes of gravitons and scalar particles was studied back in 1960s [12–15]. The original soft graviton theorem corresponding to some scattering process with a soft graviton attached to one of it's legs, published by Steven Weinberg in 1965 [1], was claiming that the resulting amplitude with soft graviton differs from the original one just by multiplying it by some factor depending on the particles' momenta with the $1/q$ dependence on the soft graviton momenta plus terms $\mathcal{O}(1)$ of it. In 2014 Cachazo and Strominger [16] extended the theorem for subleading order, calculating the second term in the soft expansion.

The soft theorem will be derived from Einstein-Hilbert action coupled with a massless scalar field ϕ:

$$S = -\int d^4x\sqrt{-g}\left[\frac{2}{\kappa^2}R + \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi\right], \tag{22}$$

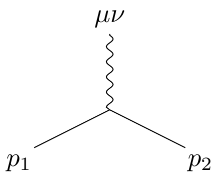
in the weak field expansion $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$. Imposing harmonic gauge $\partial^\mu h_{\mu\nu} = \frac{1}{2}\partial_\nu h$, one can expand the gravitational and scalar parts of the Lagrangian as

$$\begin{aligned}
\mathcal{L}_{\mathrm{g}} &= -\frac{2}{\kappa^2}R - \frac{1}{2}\partial_\sigma h_{\mu\nu}\partial^\sigma h^{\mu\nu} + \frac{1}{2}\partial_\mu h\partial^\mu h + \partial^\mu h_{\mu\nu}\partial_\rho h^{\nu\rho} - \partial_\mu h^{\mu\nu}\partial_\nu h + \cdots, \\
\mathcal{L}_s &= -\frac{1}{2}\sqrt{-g}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi = -\frac{1}{2}\partial^\mu\phi\partial_\mu\phi + \frac{1}{2}\kappa h^{\mu\nu}\left[\partial_\mu\phi\partial_\nu\phi - \frac{1}{2}\eta_{\mu\nu}\eta^{\sigma\rho}\partial_\sigma\phi\partial_\rho\phi\right] + \cdots.
\end{aligned} \tag{23}$$

and find the Feynman rules for such theory [17]:

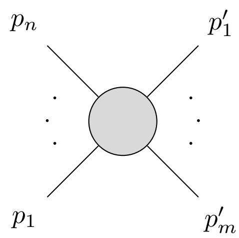
$$\underset{p}{\text{———}} = \frac{-i}{p^2 - i\varepsilon}, \tag{24}$$

$$\mu\nu \underset{q}{\sim\!\sim\!\sim} \alpha\beta = \frac{-i}{2}\frac{\eta_{\mu\alpha}\eta_{\nu\beta} + \eta_{\mu\beta}\eta_{\nu\alpha} - \eta_{\mu\nu}\eta_{\alpha\beta}}{p^2 - i\varepsilon}, \tag{25}$$



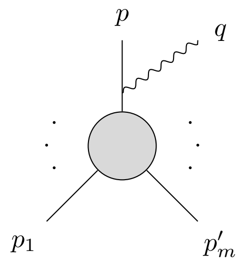
$$= \frac{i\kappa}{2}(p_{1\mu}p_{2\nu} + p_{1\nu}p_{2\mu} - \eta_{\mu\nu}p_1\cdot p_2). \tag{26}$$

Consider a scattering process with n incoming particles of momenta $\{p_1, p_2, \ldots, p_n\}$ and m outgoing particles of momenta $\{p'_1, p'_2, \ldots, p'_m\}$:



(27)

One then add an outgoing soft (low-energy) graviton with momentum q, attached to the external line of the process Feynman diagram with momentum $p \in \{p_1, p_2, \ldots, p_n, p'_1, p'_2, \ldots, p'_m\}$.



(28)

This will give the original scattering amplitude an additional factor consisted of

- propagator with momentum $p + q$ (or $p - q$ depending on is the initial particle incoming or outgoing):

$$\frac{-i}{(2\pi)^4}\frac{1}{(p + \eta q)^2 + m^2 - i\epsilon}, \tag{29}$$

  where $\eta = \pm 1$ shows was the particle incoming or outgoing,

- vertex $p + q \to p$ (or $p \to p - q$ in case of an incoming particle):

$$\frac{i}{2}(2\pi)^4\sqrt{8\pi G}(2p^\mu+\eta q^\mu)(2p^\nu+\eta q^\nu). \tag{30}$$

For $q\to 0$ the resulting factor will be written as

$$\begin{aligned}
&\frac{-i}{(2\pi)^4}\frac{1}{(p+\eta q)^2+m^2-i\epsilon}\times\frac{i}{2}(2\pi)^4\sqrt{8\pi G}(2p^\mu+\eta q^\mu)(2p^\nu+\eta q^\nu)=\\
&=\frac{\sqrt{8\pi G}}{2}\frac{4p^\mu p^\nu+2\eta(p^\mu q^\nu+q^\mu p^\nu)+q^\mu q^\nu}{p^2+2\eta p\cdot q+q^2+m^2-i\epsilon}=\Big[p^2+m^2=0\Big]=\\
&=\frac{\sqrt{8\pi G}}{2}\frac{4p^\mu p^\nu}{2\eta p\cdot q-i\epsilon}+O(1)=\sqrt{8\pi G}\frac{\eta p^\mu p^\nu}{p\cdot q-i\eta\epsilon}+O(1).
\end{aligned} \tag{31}$$

This is the leading order contribution of the 1 soft graviton scattering with a $1/(p\cdot q)$ pole. Now, if attaching this soft graviton to an arbitary diagram, one has to sum this result (31) over all external lines, which will be given as:

$$\sum_{i=1}^{m+n}\sqrt{8\pi G}\frac{\eta p^\mu p^\nu}{p\cdot q-i\eta\epsilon}=\sqrt{8\pi G}\Big(\sum_{i=1}^{m}\frac{p_i'^\mu p_i'^\nu}{p_i'\cdot q-i\epsilon}-\sum_{i=1}^{n}\frac{p_i^\mu p_i^\nu}{p_i\cdot q+i\epsilon}\Big)\equiv\sqrt{8\pi G}S^{(0)\mu\nu}. \tag{32}$$

This way, the amplitude for a scattering process with one soft graviton ($q\to 0$), originally obtained in [1], can be constructed as follows, provided that the original $m+n$-point amplitude (27) is $\mathcal{A}_{m+n}$:

$$\mathcal{A}_{m+n+1\text{ s.g.}}(p_1,\dots p_m',q)=\sqrt{8\pi G}S^{(0)}_{\mu\nu}\epsilon^{\mu\nu}\mathcal{A}_{m+n}(p_1,\dots p_m')+O(1). \tag{33}$$

If one is interested in the subleading order $O(1)$ of soft graviton theorem, it was shown in [16] that

$$\mathcal{A}_{n+m+1}(p_1,\dots p_m',q)=\left(S^{(0)}+S^{(1)}\right)\mathcal{A}_{n+m}(p_1,\dots p_m',q)+\mathcal{O}(q^1), \tag{34}$$

with subleading soft factor

$$S^{(1)}\equiv -i\sum_{a=1}^{m+n}\frac{E_{\mu\nu}k_a^\mu\,(q_\rho J_a^{\rho\nu})}{q\cdot k_a}. \tag{35}$$

Here arises the angular momentum factor, which was absent in the leading order.

1.1.3 Memory effect

Now we will move to describing gravitational memory effect, the final part of IR triangle. The memory effect characterizes a pair of inertial detectors stationed near $\mathscr{I}^+$ in a region with no Bondi news $N_{zz}=M_{zz}=0$ at both late and early times. At intermediate times, gravity waves pass through, causing oscillating distortions in their relative separation. This relative separation lies in a plane $(z,\bar z)$ perpendicular to the direction of gravitational wave propagation, and we will denote the corresponding positions as $(s^z,s^{\bar z})$.

As the gravitational wave propagates perpendicular to the plane of two inertial masses (detectors), it is displaced. After the gravitational wave has passed, the masses are permanently displaced, due to the memory effect.

The equation of geodesic deviation implies

$$r^2\gamma_{z\bar z}\partial_u^2 s^{\bar z}=-R_{uzuz}s^z, \tag{36}$$

where

$$R_{uzuz}=-\frac{r}{2}\partial_u^2 C_{zz}. \tag{37}$$

Now we just integrate this equation and obtain the shift:

$$\Delta s^{\bar z}=\frac{\gamma^{z\bar z}}{2r}\Delta C_{zz}s^z. \tag{38}$$

This is the gravitational memory effect. The difference ΔC_{zz} between initial and final transverse metric components need not vanish, as flatness does not require $C_{zz}=0$. Proposals to measure the gravitational memory effect with a variety of methods are given in [18,19].

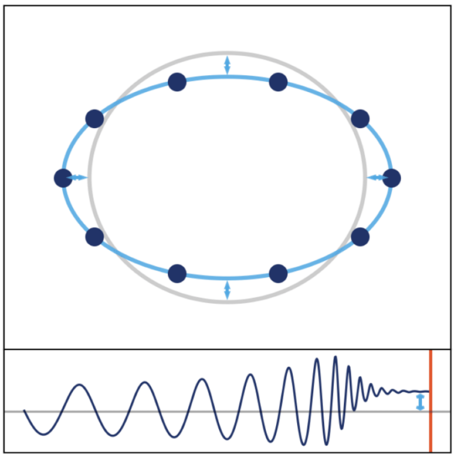


Figure 3: The top image shows how the circle of probe masses (detectors) is displaced before (grey) and after (blue) gravitational wave passes. The bottom graph depicts how the distance between two of the probe masses (detectors) starts oscillating when the gravitational wave passes through and ends up being permanently changed by some distance.

1.1.4 Soft Theorem $\leftrightarrow$ Asymptotic Symmetry (Ward identities)

Before deriving Ward identities from soft theorem result, some time will be dedicated to understanding what is a Ward identity, which follows from supertranslation symmetry of S-matrix

$$Q_f^+ S - S Q_f^- = 0. \tag{39}$$

Firstly, one should fix some important notations. In- and out- states on the conformal sphere at past and future null infinities ($\mathcal{I}^\pm$) will be denoted as $|z_1^{\text{in}}, \cdots\rangle$ and $\langle z_1^{\text{out}}, \cdots|$, respectively .

Approaching point of $u, r \to +\infty$, the parametrization of spatial momentum is given by

$$\vec{p} = \frac{\omega}{1+z\bar{z}}(z+\bar{z}, -iz+i\bar{z}, 1-z\bar{z}), \quad \vec{p}\cdot\vec{p} = \omega^2. \tag{40}$$

Asymptotical approximation of gravitational field at $u, r \to +\infty$ can be written in terms of mode expansion as

$$h_{\mu\nu}^{\text{out}}(x) = \sum_{\alpha=\pm} \int \frac{d^3q}{(2\pi)^3 2\omega_q} \left[\varepsilon_{\mu\nu}^{\alpha*}(\vec{q}) a_\alpha^{\text{out}}(\vec{q}) e^{iq\cdot x} + \varepsilon_{\mu\nu}^{\alpha}(\vec{q}) a_\alpha^{\text{out}\,\dagger}(\vec{q}) e^{-iq\cdot x}\right], \tag{41}$$

where $q^0 = \omega_q = |\vec{q}|$, $\alpha = \pm$ are the two possible graviton helicities and the creation-annihilation commutation relation is taken to be

$$[a_\alpha^{\text{out}}(\vec{q}), a_\beta^{\dagger\text{out}}(\vec{q}')] = \delta_{\alpha\beta}(2\omega_q)(2\pi)^3\delta^3(\vec{q}-\vec{q}'). \tag{42}$$

Inserting annihilation operator $a_\alpha^{\text{out}}(\vec{q})$ corresponds to introducing a graviton of momentum q and polarization α in terms of soft graviton theorem.

The graviton four-momentum can be parametrized as

$$q^\mu = \frac{\omega_q}{1+w\bar{w}}(1+w\bar{w}, w+\bar{w}, -i(w-\bar{w}), 1-w\bar{w}), \tag{43}$$

with polarization tensors $\varepsilon^{\pm\mu\nu} = \varepsilon^{\pm\mu}\varepsilon^{\pm\nu}$ may be written down as

$$\begin{aligned}\varepsilon^{+\mu}(\vec{q}) &= \frac{1}{\sqrt{2}}(\bar{w}, 1, -i, -\bar{w}),\\ \varepsilon^{-\mu}(\vec{q}) &= \frac{1}{\sqrt{2}}(w, 1, i, -w),\end{aligned} \tag{44}$$

obeying $\varepsilon^{\pm\mu\nu}q_\nu = \varepsilon^{\pm\mu}q_\mu = 0$ and

$$\varepsilon_z^+(\vec{q}) = \partial_z x^\mu \varepsilon_\mu^+(\vec{q}) = \frac{\sqrt{2}r\bar{z}(w-\bar{z})}{(1+z\bar{z})^2}, \quad \varepsilon_z^-(\vec{q}) = \partial_z x^\mu \varepsilon_\mu^-(\vec{q}) = \frac{\sqrt{2}r(1+w\bar{z})}{(1+z\bar{z})^2}. \tag{45}$$

From the expression for metric in Bondi coordinates near future null infinity, substituting mode expansion (41) and the definition of $h_{zz} = \partial_z x^\mu \partial_z x^\nu h_{\mu\nu}$ one can see

$$C_{zz}(u,z,\bar{z}) = \kappa \lim_{r\to\infty} \frac{1}{r} h^{\text{out}}_{zz}(r,u,z,\bar{z}) = \\ = \kappa \lim_{r\to\infty} \frac{1}{r} \partial_z x^\mu \partial_z x^\nu \sum_{\alpha=\pm} \int \frac{d^3 q}{(2\pi)^3 2\omega_q} \left[\varepsilon^{\alpha *}_{\mu\nu}(\vec{q}) a^{\text{out}}_\alpha(\vec{q}) e^{-i\omega_q u - i\omega_q r(1-\cos\theta)} + \text{h.c.} \right], \tag{46}$$

where θ is an angle between $\vec{x}$ and $\vec{q}$.

As approaching the spatial infinity $r \to \infty$, the expression under integral will oscillate with stationary points $0, \pi$, and therefore approximating over the spatial momentum the integral may be written as

$$C_{zz} = -\frac{i\kappa}{4\pi^2(1+z\bar{z})^2} \int_0^\infty d\omega_q \left[a^{\text{out}}_+(\omega_q \hat{x}) e^{-i\omega_q u} - a^{\text{out}\,\dagger}_-(\omega_q \hat{x}) e^{i\omega_q u} \right]. \tag{47}$$

Now the outgoing Bondi news tensor may be rewritten in terms of creation and annihilation operators as follows:

$$N^\omega_{zz}(z,\bar{z}) \equiv \int_{-\infty}^\infty du e^{i\omega u} \partial_u C_{zz} = \\ = -\frac{\kappa}{2\pi(1+z\bar{z})^2} \int_0^\infty d\omega_q \left[a^{\text{out}}_+(\omega_q \hat{x}) \delta(\omega_q - \omega) + a^{\text{out}}_-(\omega_q \hat{x})^\dagger \delta(\omega_q + \omega) \right]. \tag{48}$$

If $\omega > 0$ ($\omega < 0$), only the first (second) term survives and one can define

$$N^\omega_{zz}(z,\bar{z}) = -\frac{\kappa\omega a^{\text{out}}_+(\omega\hat{x})}{2\pi(1+z\bar{z})^2}, \\ N^{-\omega}_{zz}(z,\bar{z}) = -\frac{\kappa\omega a^{\text{out}}_-(\omega\hat{x})^\dagger}{2\pi(1+z\bar{z})^2}; \tag{49}$$

where $\omega > 0$.

The zero mode will be defined as

$$N^0_{zz} \equiv \lim_{\omega\to 0^+} \frac{1}{2}(N^\omega_{zz} + N^{-\omega}_{zz}) = -\frac{\kappa}{4\pi(1+z\bar{z})^2} \lim_{\omega\to 0^+} \left[\omega a^{\text{out}}_+(\omega\hat{x}) + \omega a^{\text{out}}_-(\omega\hat{x})^\dagger \right] \tag{50}$$

The same analysis holds for the past null infinity, where in the end one arrives to expression for incoming Bondi news tensor

$$M^0_{zz}(z,\bar{z}) \equiv -\frac{\kappa}{4\pi(1+z\bar{z})^2} \lim_{\omega\to 0^+} \left[\omega a^{\text{in}}_+(\omega\hat{x}) + \omega a^{\text{in}}_-(\omega\hat{x})^\dagger \right]. \tag{51}$$

Now one can define an operator

$$\mathcal{O}_{zz} \equiv N^0_{zz}(z,\bar{z}) + M^0_{zz}(z,\bar{z}), \tag{52}$$

which will arise a in the next section when deriving Ward identity, to merge Fock space and Bondi metric notations.

If one fix arbitary function $f(w,\bar{w}) = \frac{1}{z-w}$, the matrix element between such states can be then written as

$$\langle z_1^{\text{out}}, \cdots | : P_z S : | z_1^{\text{in}}, \cdots \rangle = \langle z_1^{\text{out}}, \cdots | S | z_1^{\text{in}}, \cdots \rangle \left[\sum_{k=1}^m \frac{E_k^{\text{out}}}{z - z_k^{\text{out}}} - \sum_{k=1}^n \frac{E_k^{\text{in}}}{z - z_k^{\text{in}}} \right], \tag{53}$$

where current P_z is given by

$$P_z \equiv \frac{1}{2G} \left(\int_{-\infty}^\infty dv \partial_v V_z - \int_{-\infty}^\infty du \partial_u U_z \right) = \frac{1}{4G} \gamma^{z\bar{z}} \partial_z \mathcal{O}_{zz} \tag{54}$$

with $\mathcal{O}_{zz}$ defined in (52) and for incoming and outgoing energies $E_k^{\text{in}}, E_k^{\text{out}}$ holds conservation law

$$\sum_{k=1}^m E_k^{\text{out}} = \sum_{k=1}^n E_k^{\text{in}} \tag{55}$$

Ward identity, corresponding to supertranslations, is given by expression (53).

Now, after understanding how supertranslation symmetry of asumptotically flat spacetimes is leading to Ward identity, one may be interested in finding connection between it and soft theorem. To do so, some preparation may be needed.

First of all it is useful to write down momenta and polarization vectors in holomorphic coordinates::

$$
\begin{aligned}
p_k^\mu &= E_k^{\mathrm{in}}\left(1, \frac{z_k^{\mathrm{in}}+\bar{z}_k^{\mathrm{in}}}{1+z_k^{\mathrm{in}}\bar{z}_k^{\mathrm{in}}}, \frac{-i\left(z_k^{\mathrm{in}}-\bar{z}_k^{\mathrm{in}}\right)}{1+z_k^{\mathrm{in}}\bar{z}_k^{\mathrm{in}}}, \frac{1-z_k^{\mathrm{in}}\bar{z}_k^{\mathrm{in}}}{1+z_k^{\mathrm{in}}\bar{z}_k^{\mathrm{in}}}\right), \\
p_k'^\mu &= E_k^{\mathrm{out}}\left(1, \frac{z_k^{\mathrm{out}}+\bar{z}_k^{\mathrm{out}}}{1+z_k^{\mathrm{out}}\bar{z}_k^{\mathrm{out}}}, \frac{-i\left(z_k^{\mathrm{out}}-\bar{z}_k^{\mathrm{out}}\right)}{1+z_k^{\mathrm{out}}\bar{z}_k^{\mathrm{out}}}, \frac{1-z_k^{\mathrm{out}}\bar{z}_k^{\mathrm{out}}}{1+z_k^{\mathrm{out}}\bar{z}_k^{\mathrm{out}}}\right), \\
q^\mu &= \omega\left(1, \frac{z+\bar{z}}{1+z\bar{z}}, \frac{-i\left(z-\bar{z}\right)}{1+z\bar{z}}, \frac{1-z\bar{z}}{1+z\bar{z}}\right), \\
\varepsilon^{+\mu}(q) &= \frac{1}{\sqrt{2}}\left(\bar{z}, 1, -i, -\bar{z}\right),
\end{aligned}
\tag{56}
$$

Taking into account (52), the S-matrix element may be written as

$$
\begin{aligned}
\langle z_1^{\mathrm{out}}, \cdots | : \mathcal{O}_{zz} S : | z_1^{\mathrm{in}}, \cdots \rangle = -\frac{\kappa}{4\pi(1+z\bar{z})^2} \lim_{\omega\to 0^+} \Big[&\omega \langle z_1^{\mathrm{out}}, \cdots | a_+^{\mathrm{out}}(\omega\hat{x}) S | z_1^{\mathrm{in}}, \cdots \rangle \\
&+ \omega \langle z_1^{\mathrm{out}}, \cdots | S a_-^{\mathrm{in}}(\omega\hat{x})^\dagger | z_1^{\mathrm{in}}, \cdots \rangle \Big].
\end{aligned}
\tag{57}
$$

Here, the fact that $a_-^{\mathrm{out}}(\omega\hat{x})^\dagger$ $(a_+^{\mathrm{in}}(\omega\hat{x}))$ annihilates the out (in) state for $\omega \to 0$ was used. The first term corresponds to an outgoing positive helicity soft graviton, while the second – to an incoming negative helicity, both with spatial momenta $\omega\hat{x}$. Therefore, the sum of two amplitudes can be combined in one term

$$
\langle z_1^{\mathrm{out}}, \cdots | : \mathcal{O}_{zz} S : | z_1^{\mathrm{in}}, \cdots \rangle = -\frac{\kappa}{2\pi(1+z\bar{z})^2} \lim_{\omega\to 0^+} \left[\omega \langle z_1^{\mathrm{out}}, \cdots | a_+^{\mathrm{out}}(\omega\hat{x}) S | z_1^{\mathrm{in}}, \cdots \rangle \right].
\tag{58}
$$

Writing now the soft graviton theorem (32) with a positive helicity outgoing graviton

$$
\begin{aligned}
&\lim_{\omega\to 0} \left[\omega \langle z_1^{\mathrm{out}}, \cdots | a_+^{\mathrm{out}}(\vec{q}) S | z_1^{\mathrm{in}}, \cdots \rangle \right] = \\
&= \frac{\kappa}{2} \lim_{\omega\to 0^+} \left[\sum_{k=1}^{m} \frac{\omega [p_k' \cdot \varepsilon^+(q)]^2}{p_k' \cdot q} - \sum_{k=1}^{n} \frac{\omega [p_k \cdot \varepsilon^+(q)]^2}{p_k \cdot q} \right] \langle z_1^{\mathrm{out}}, \cdots | S | z_1^{\mathrm{in}}, \cdots \rangle.
\end{aligned}
\tag{59}
$$

Substituting (59) into (58) and using coordinates (56), one finds the final expression for S-matrix element with insertion of $\mathcal{O}_{zz}$:

$$
\begin{aligned}
\langle z_1^{\mathrm{out}}, \cdots | : \mathcal{O}_{zz} S : | z_1^{\mathrm{in}}, \cdots \rangle = &\frac{8G}{(1+z\bar{z})^2} \langle z_1^{\mathrm{out}}, \cdots | S | z_1^{\mathrm{in}}, \cdots \rangle \\
&\times \left[\sum_{k=1}^{m} \frac{E_k^{\mathrm{out}}(\bar{z} - \bar{z}_k^{\mathrm{out}})}{(z - z_k^{\mathrm{out}})(1 + z_k^{\mathrm{out}}\bar{z})} - \sum_{k=1}^{n} \frac{E_k^{\mathrm{in}}(\bar{z} - \bar{z}_k^{\mathrm{in}})}{(z - z_k^{\mathrm{in}})(1 + \bar{z}_k^{\mathrm{in}}\bar{z})} \right].
\end{aligned}
\tag{60}
$$

Now, to reconnect with Ward identities, one should remember that $P_z = \frac{1}{4G}\gamma^{z\bar{z}}\partial_z \mathcal{O}_{zz}$ and calculate matrix element with P_z inserted:

$$
\begin{aligned}
\langle z_1^{\mathrm{out}}, \cdots | : P_z S : | z_1^{\mathrm{in}}, \cdots \rangle &= \frac{1}{4G} \gamma^{z\bar{z}} \partial_{\bar{z}} \langle z_1^{\mathrm{out}}, \cdots | : \mathcal{O}_{zz} S : | z_1^{\mathrm{in}}, \cdots \rangle \\
&= \langle z_1^{\mathrm{out}}, \cdots | S | z_1^{\mathrm{in}}, \cdots \rangle \left[\sum_{k=1}^{m} \frac{E_k^{\mathrm{out}}}{z - z_k^{\mathrm{out}}} - \sum_{k=1}^{n} \frac{E_k^{\mathrm{in}}}{z - z_k^{\mathrm{in}}} \right] \\
&+ \langle z_1^{\mathrm{out}}, \cdots | S | z_1^{\mathrm{in}}, \cdots \rangle \left[\sum_{k=1}^{m} \frac{E_k^{\mathrm{out}} z_k^{\mathrm{out}}}{1 + z_k^{\mathrm{out}}\bar{z}} - \sum_{k=1}^{n} \frac{E_k^{\mathrm{in}} z_k^{\mathrm{in}}}{1 + z_k^{\mathrm{in}}\bar{z}} \right].
\end{aligned}
\tag{61}
$$

In this expression the last term vanishes due to momentum conservation and what left is nothing else but the Ward identity

$$\langle z_1^{\text{out}}, \cdots | : P_z S : | z_1^{\text{in}}, \cdots \rangle = \langle z_1^{\text{out}}, \cdots | S | z_1^{\text{in}}, \cdots \rangle \left[\sum_{k=1}^{m} \frac{E_k^{\text{out}}}{z - z_k^{\text{out}}} - \sum_{k=1}^{n} \frac{E_k^{\text{in}}}{z - z_k^{\text{in}}} \right], \tag{62}$$

which was obtained using the machinery of quantum field theory approach.

1.1.5 Memory Effect $\leftrightarrow$ Soft Theorem (Fourier Transform)

Recall that the Bondi news tensor is defined as the proper time derivative of the shear: $N_{zz} = \partial_u C_{zz}$. The total permanent shift in the metric is just the integral of the news over all time:

$$\Delta C_{zz} = \int_{-\infty}^{\infty} du \, \partial_u C_{zz} = \int_{-\infty}^{\infty} du \, N_{zz}(u) \,. \tag{63}$$

But this integral is exactly the zero-frequency limit of the Fourier transform of the news tensor!

$$\Delta C_{zz} = \lim_{\omega \to 0} \int_{-\infty}^{\infty} du \, e^{i\omega u} N_{zz}(u) \equiv N_{zz}^{\omega=0} \,. \tag{64}$$

Looking back at our equation (50), we see that the zero-frequency news N_{zz}^0 is exactly the zero-energy limit of the graviton annihilation operator $\lim_{\omega \to 0} \omega a^{out}(\omega)$. Thus, the classical memory shift ΔC_{zz} is mathematically equivalent to the insertion of a soft graviton.

1.1.6 Memory Effect $\leftrightarrow$ Asymptotic Symmetry (Vacuum Transition)

To complete the IR triangle, we need to understand how does the shear C_{zz} transform under a BMS supertranslation $f(z, \bar{z})$. We should act with a Lie derivative on a corresponding component of metric tensor $\delta g_{zz} = \mathcal{L}_\xi g_{zz}$. As a result we will obtain the following constraint on the shear tensor:

$$\delta_f C_{zz} = -2D_z^2 f. \tag{65}$$

If we consider the spacetime before the wave passes ($u \to -\infty$) and after it passes ($u \to +\infty$), both regions are empty Minkowski vacua (news $N_{zz} = 0$). However, the passing wave creates a permanent shift ΔC_{zz}. This implies that the initial and final vacua are not the same; they differ precisely by a BMS supertranslation with parameter f, such that:

$$\Delta C_{zz} = -2D_z^2 f \,. \tag{66}$$

Therefore, the memory effect is a physical, measurable transition between two degenerate Minkowski vacua related by a supertranslation.

1.2 2D CFT basics

To appreciate all the beauty of celestial holography program, we need to get to know some basics of conformal field theory. This conformal field theory will be considered on a 2-dimensional celestial sphere mentioned above, and therefore, we will restrict ourselves to introduce the basics of 2D CFT.

1.2.1 Conformal transformations in D dimensions

Definition 3. *By denoting the metric as $g_{\mu\nu}(x)$ we say that a coordinate transformation $x \to x'$ is* ***conformal*** *if the metric is scale-invariant:*

$$g'_{\mu\nu}(x') = \Lambda(x) g_{\mu\nu}(x). \tag{67}$$

The word *conformal* arises from the fact that conformal transformations preserve angles between curves. To see which transformations can be conformal, we should look in the infinitesimal one $x'^\mu = x^\mu + \epsilon^\mu(x)$ and see how metric changes up to first order in ϵ:

$$g'_{\mu\nu}(x') = \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\lambda}{\partial x'^\nu} g_{\rho\lambda}(x) = (\delta_\mu^\rho - \partial_\mu \epsilon^\rho)(\delta_\nu^\lambda - \partial_\nu \epsilon^\lambda) g_{\rho\lambda} = g_{\mu\nu} - (\partial_\nu \epsilon_\mu + \partial_\mu \epsilon_\nu) \tag{68}$$

where we have used that $\epsilon^\mu(x) = \epsilon^\mu(x' - \epsilon(x')) = \epsilon^\mu(x') - \epsilon^\nu(x') \partial_\nu \epsilon^\mu(x') + o(\epsilon^2)$. From here we can see that the infinitesimal transformations satisfy:

$$\partial_\nu \epsilon_\mu + \partial_\mu \epsilon_\nu = (1 - \Lambda(x)) g_{\mu\nu}. \tag{69}$$

Taking now the trace of this expression, we find the form of an infinitesimal function $(1-\Lambda(x)) = \frac{2}{d}\partial_\mu\epsilon^\mu \equiv f(x)$.

Now we apply one more derivative to the equation and for simplicity consider flat spacetime $g_{\mu\nu} = \eta_{\mu\nu}$:

$$\partial_\rho\partial_\nu\epsilon_\mu + \partial_\rho\partial_\mu\epsilon_\nu = \partial_\rho f\eta_{\mu\nu}, \tag{70}$$
$$\partial_\nu\partial_\mu\epsilon_\rho + \partial_\nu\partial_\rho\epsilon_\mu = \partial_\nu f\eta_{\mu\rho}, \tag{71}$$
$$\partial_\mu\partial_\rho\epsilon_\nu + \partial_\mu\partial_\nu\epsilon_\rho = \partial_\mu f\eta_{\nu\rho}. \tag{72}$$

If we now sum (70)+(71)-(72) and then contract with $\eta^{\nu\rho}$, we arrive to an equation:

$$2\partial^2\epsilon_\mu = (2-d)\partial_\mu f. \tag{73}$$

Taking now derivative ∂^μ from both sides and remembering $f(x) = \frac{2}{d}\partial_\mu\epsilon^\mu$, we arrive to

$$(d-1)\partial^2 f = 0. \tag{74}$$

Solving in for $d \geq 3$ we obtain the following types of conformal transformations with their infinitesimal generators:

$$\text{(translations)} \qquad x'^\mu = x^\mu + a^\mu, \qquad P_\mu - i\partial_\mu; \tag{75}$$
$$\text{(dilations)} \qquad x'^\mu = \alpha x^\mu, \qquad D = -ix^\mu\partial_\mu; \tag{76}$$
$$\text{(rigid rotations)} \qquad x'^\mu = M^\mu_{\ \nu}x^\nu, \qquad L_{\mu\nu} = i(x_\mu\partial_\nu - x_\nu\partial_\mu); \tag{77}$$
$$\text{(SCT)} \qquad x'^\mu = \frac{x^\mu - b^\mu x^2}{1 - 2b\cdot x + b^2x^2}, \qquad K_\mu = -i(2x_\mu x^\nu\partial_\nu - x^2\partial_\mu). \tag{78}$$

However, the most interesting case for us, namely $d = 2$, will be considered in the following.

1.2.2 Conformal transformations in 2 dimensions

The case of 2D is of special interest for us because as we have seen, the celestial sphere is 2-dimensional. Therefore, to ask for conformal symmetry on a celestial sphere, we need to reduce our search to $D = 2$.

Let's consider a plane parametrized by (z^0, z^1) and a change of coordinates $z \to w$, under which the metric tensor transforms as follows:

$$g'^{\mu\nu}(w) = \frac{\partial w^\mu}{\partial z^\alpha}\frac{\partial w^\nu}{\partial z^\beta}g^{\alpha\beta}(z). \tag{79}$$

If this coordinate is conformal, then after applying it (since the original metric is flat) the conditions (67) for the components g'^{00} and g'^{11}, g'^{01} and g'^{10} become:

$$\left(\frac{\partial w^0}{\partial z^0}\right)^2 + \left(\frac{\partial w^0}{\partial z^1}\right)^2 = \left(\frac{\partial w^1}{\partial z^0}\right)^2 + \left(\frac{\partial w^1}{\partial z^1}\right)^2, \tag{80}$$
$$\frac{\partial w^0}{\partial z^0}\frac{\partial w^1}{\partial z^0} + \frac{\partial w^0}{\partial z^1}\frac{\partial w^1}{\partial z^1} = 0.$$

When solving this w.r.t. $\frac{\partial w^0}{\partial z^0}, \frac{\partial w^0}{\partial z^1}, \frac{\partial w^1}{\partial z^0}, \frac{\partial w^1}{\partial z^1}$, we obtain the Cauchy-Rieman conditions for holomorhic

$$\frac{\partial w^0}{\partial z^1} = \frac{\partial w^1}{\partial z^0}, \quad \frac{\partial w^0}{\partial z^0} = -\frac{\partial w^1}{\partial z^1}, \tag{81}$$

or antiholomoprhic functions

$$\frac{\partial w^0}{\partial z^1} = -\frac{\partial w^1}{\partial z^0}, \quad \frac{\partial w^0}{\partial z^0} = \frac{\partial w^1}{\partial z^1}. \tag{82}$$

This motivates us to use holomorphic and antiholomorphic coordinates $(z, \bar{z})$ from now on:

$$z = z^0 + iz^1, \tag{83}$$
$$\bar{z} = z^0 - iz^1, \tag{84}$$
$$\partial_z = \frac{1}{2}(\partial_0 - i\partial_1) \equiv \partial, \tag{85}$$
$$\partial_{\bar{z}} = \frac{1}{2}(\partial_0 + i\partial_1) \equiv \bar{\partial}. \tag{86}$$

In these coordinates we rewrite the metric tensor as:

$$g_{\mu\nu} = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}, \quad g^{\mu\nu} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}. \tag{87}$$

In this language, conditions (80) become just $\bar{\partial} w = 0$, which means conformal transformations are just holomorphic (or analytic) functions on the complex plane $f(z)$.

Important remark here is to remember that originally (z^0, z^1) were real independent coordinates in $\mathbb{R}^2$, and therefore $(z, \bar{z})$ in principle should be considered as two dependent complex coordinates. In practice, we say that we treat z and $\bar{z}$ as two independent complex variables on $\mathbb{C}^2$, and z^0, z^1 therefore are complexified. However, when we return back to our physical space, we should imply condition $\bar{z} = z^*$, returning to our real surface.

1.2.3 Global conformal transformations

The Cauchy-Riemann conditions are local conditions, which in general can mean that $f(z)$ is not well-defined everywhere, but only in some area. Therefore, we will distinct global conformal transformations that are well-defined on the whole Riemann sphere and local ones that are not.

The global conformal transformations (a.k.a. Moebius or projective transformations)

$$f(z) = \frac{az + b}{cz + d}, \quad ad - bc = 1, \quad a, b, c, d \in \mathbb{C}, \tag{88}$$

form a $SL(2, \mathbb{C})$ group of 2x2 matrices $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with unit determinant.

A composition of two Moebius transformations is indeed another one $f_2 \circ f_1$ and is given by matrix multiplication of corresponding matrices $A_1 A_2$, therefore forming a group.

1.2.4 Conformal generators

We now turn our attention to the local conformal group. Any infinitesimal holomorphic transformatino can be expressed as following:

$$z' = z + \epsilon(z), \quad \epsilon(z) = \sum_{-\infty}^{\infty} c_n z^{n+1}. \tag{89}$$

Here the transformation admits a Laurent expansion around zero, but is a holomorphic function in a ring around zero. Therefore, to see which generators induce such mapping, we should expand up to first order in ϵ a scalar (for simplicity) field $\phi(z, \bar{z})$ living on the plane after an infinitesimal transformation:

$$\phi'(z', \bar{z}') = \phi(z, \bar{z}) = \phi(z', \bar{z}') - \epsilon(z')\partial'\phi(z', \bar{z}') - \bar{\epsilon}(\bar{z}')\bar{\partial}'\phi(z', \bar{z}'), \tag{90}$$

$$\delta\phi = -\epsilon(z)\partial\phi - \bar{\epsilon}(\bar{z})\bar{\partial}\phi = \sum_n \left[c_n l_n \phi(z, \bar{z}) + \bar{c}_n \bar{l}_n \phi(z, \bar{z}) \right], \tag{91}$$

where $l_n, \bar{l}_n$ are nothing else but generators of conformal algebra

$$l_n = -z^{n+1}\partial_z, \bar{l}_n = -\bar{z}^{n+1}\partial_{\bar{z}}, \tag{92}$$

obeying commutation relations:

$$[l_m, l_n] = (m - n) l_{m+n}, \tag{93}$$

$$[\bar{l}_m, \bar{l}_n] = (m - n) \bar{l}_{m+n}, \tag{94}$$

$$[l_m . \bar{l}_n] = 0. \tag{95}$$

Both l_n and $\bar{l}_n$ obey Witt algebra, so sometimes we will say that algebra of conformal generators is nothing else but two copies of Witt algebra: Witt$\otimes$Witt.

One may notice that generators l_{-1}, l_0, l_1 form subalgebra which reproduces the global conformal algebra $sl(2, \mathbb{C})$. Notice also that when $n = -1, 0, 1$, the generators are holomorphic on the whole Riemann sphere.

Exercise 3. *Demanding that the generators should be holomorphic at* $z = 0$ *and* $z = \infty$ *(make an inversion* $z = \lim_{w \to 0} \frac{1}{w}$*) show that* $n \in \{-1, 0, 1\}$.

For some intuition we will list some of the generators and their action below:

- $l_{-1} = -\partial_z$ – translations on the plane;
- $l_0 = -z\partial_z$ – scale transformations + rotations;
- $l_1 = -z^2\partial_z$ – SCT (special conformal transformations);
- $l_n + \bar{l}_n, i(l_n - \bar{l}_n)$ – generators that preserve the real surface $z^0, z^1 \in \mathbb{R}$.

1.2.5 Primary fields

To build a space of states in 2D CFT, we need a specific set of local fields called primary fields. These fields are annihilated by lowering generators of conformal algebra and have a special transformation rule they obey. All the other fields in CFT can be obtained by acting with raising generators on the primary fields and are called their descendants.

Before defining a primary field, we should introduce some important notation:

Definition 4. *For a field with scaling dimension Δ and spin s we define $h = \frac{1}{2}(\Delta + s)$ to be his* ***holomorphic conformal dimension****, while $\bar{h} = \frac{1}{2}(\Delta - s)$ an* ***antiholomorphic conformal dimension****.*

And now we can define the primary fields:

Definition 5. *A field that under a conformal map $z \to w(z), \bar{z} \to \bar{w}(\bar{z})$ transforms as*

$$\phi'(w, \bar{w}) = \left(\frac{dw}{dz}\right)^{-h} \left(\frac{d\bar{w}}{d\bar{z}}\right)^{-\bar{h}} \phi(z, \bar{z})$$

is called a ***quasi-primary field****.*

Notice that a quasi-primary field can transform this way under global $SL(2, \mathbb{C})$ transformations only. When a field transforms according to this rule under *any* local conformal transformation, we call it a ***primary*** field.

Rigorosity Alert 1. *Now we should make an important note: even though it would be mathematically correct, in the latter we will not use the word "quasi-primary", when talking about fields transforming this way under global $SL(2, \mathbb{C})$ transformations. We will start calling these fields primaries, and later on we will discover that they are primaries indeed if we consider superrotation symmetry.*

1.2.6 Correlation functions

We should also quickly mention which symmetries and properties do correlation functions have in CFT.

We start with a two-point function for a quasi-primary field with action $S[\phi]$ (using complex coordinates $(z_i, \bar{z}_i)$):

$$\langle \phi_1(z_1, \bar{z}_1)\phi_2(z_2, \bar{z}_2)\rangle = \frac{1}{Z}\int [d\Phi]\phi_1(z_1, \bar{z}_1)\phi_2(z_2, \bar{z}_2) \exp\{-S[\Phi]\}. \tag{96}$$

The conformal invariance of the action and of the functional integration measure leads to the following transformation of the correlation function $z \to w(z), \bar{z} \to \bar{w}(\bar{z})$:

$$\langle \phi_1(z_1, \bar{z}_1)\phi_2(z_2, \bar{z}_2)\rangle = \left(\frac{dw}{dz}\right)^{h_1}_{z_1} \left(\frac{d\bar{w}}{d\bar{z}}\right)^{\bar{h}_1}_{\bar{z}_1} \left(\frac{dw}{dz}\right)^{h_2}_{z_2} \left(\frac{d\bar{w}}{d\bar{z}}\right)^{\bar{h}_2}_{\bar{z}_2} \langle \phi_1(w_1, \bar{w}_1)\phi_2(w_2, \bar{w}_2)\rangle. \tag{97}$$

If we specialize to a scale and rotation transformation $z \to \lambda z$, $\bar{z} \to \bar{\lambda}\bar{z}$ we obtain

$$\langle \phi_1(z_1, \bar{z}_1)\phi_2(z_2, \bar{z}_2)\rangle = \lambda^{h_1+h_2}\bar{\lambda}^{\bar{h}_1+\bar{h}_2}\langle \phi_1(\lambda z_1, \bar{\lambda}\bar{z}_1)\phi_2(\lambda z_2, \bar{\lambda}\bar{z}_2)\rangle. \tag{98}$$

Meanwhile translation invariance requires that the correlator depends only on the differences $z_{12} = z_1 - z_2$ and $\bar{z}_{12} = \bar{z}_1 - \bar{z}_2$. Taking into account the scaling and rotation invariance, we get:

$$\langle \phi_1(z_1, \bar{z}_1)\phi_2(z_2, \bar{z}_2)\rangle = \frac{C_{12}}{z_{12}^{h_1+h_2}\bar{z}_{12}^{\bar{h}_1+\bar{h}_2}}, \tag{99}$$

where C_{12} is a constant coefficient. It only remains to use the invariance under special conformal transformations. We recall that, for such a transformation $w(z) = \frac{z}{1-bz}$, the derivative is:

$$\frac{dw}{dz} = \frac{1}{(1-bz)^2}, \quad \frac{d\bar{w}}{d\bar{z}} = \frac{1}{(1-\bar{b}\bar{z})^2}. \tag{100}$$

The transformation for the distance $w_{12} = w_1 - w_2$ is given by:

$$w_{12} = \frac{z_{12}}{(1-bz_1)(1-bz_2)}, \quad \bar{w}_{12} = \frac{\bar{z}_{12}}{(1-\bar{b}\bar{z}_1)(1-\bar{b}\bar{z}_2)}, \tag{101}$$

therefore the covariance of the correlation function implies

$$\frac{C_{12}}{z_{12}^{h_1+h_2}\bar{z}_{12}^{\bar{h}_1+\bar{h}_2}} = \frac{C_{12}}{\gamma_1^{h_1}\bar{\gamma}_1^{\bar{h}_1}\gamma_2^{h_2}\bar{\gamma}_2^{\bar{h}_2}} \frac{(\gamma_1\gamma_2)^{h_1+h_2}(\bar{\gamma}_1\bar{\gamma}_2)^{\bar{h}_1+\bar{h}_2}}{z_{12}^{h_1+h_2}\bar{z}_{12}^{\bar{h}_1+\bar{h}_2}}, \tag{102}$$

with

$$\gamma_i = (1-bz_i), \quad \bar{\gamma}_i = (1-\bar{b}\bar{z}_i). \tag{103}$$

This constraint is identically satisfied only if $h_1 = h_2 = h$ and $\bar{h}_1 = \bar{h}_2 = \bar{h}$. In other words, two quasi-primary fields are correlated only if they have the same conformal weights:

$$\langle\phi_1\phi_2\rangle = \begin{cases} \dfrac{C_{12}}{z_{12}^{2h}\,\bar{z}_{12}^{2\bar{h}}} & \text{if } h_1 = h_2 = h \text{ and } \bar{h}_1 = \bar{h}_2 = \bar{h} \\ 0 & \text{otherwise} \end{cases}. \tag{104}$$

Performing similar analysis, denoting $z_{ij} = z_i - z_j$ and $\bar{z}_{ij} = \bar{z}_i - \bar{z}_j$, we can obtain the expression for a 3-point function:

$$\langle\phi_1\phi_2\phi_3\rangle = \frac{C_{123}}{z_{12}^{h_1+h_2-h_3}\,z_{23}^{h_2+h_3-h_1}\,z_{13}^{h_3+h_1-h_2}} \times \frac{1}{\bar{z}_{12}^{\bar{h}_1+\bar{h}_2-\bar{h}_3}\,\bar{z}_{23}^{\bar{h}_2+\bar{h}_3-\bar{h}_1}\,\bar{z}_{13}^{\bar{h}_3+\bar{h}_1-\bar{h}_2}} \tag{105}$$

Exercise 4. *Perform a similar analysis and obtain 3-point function.*

Unfortunately, conformal covariance can only carry us so far. When considering the 4-point function, we discover that the form of it might depend arbitarily on cross-ratios. There is a finite number of linearly independent cross-ratios that one can construct in 2D CFT:

$$\eta = \frac{z_{12}z_{34}}{z_{13}z_{24}}, \qquad 1-\eta = \frac{z_{14}z_{23}}{z_{13}z_{24}}, \qquad \frac{\eta}{1-\eta} = \frac{z_{12}z_{34}}{z_{14}z_{23}}, \tag{106}$$

but unfortunately an infinite number of real functions $f(\eta,\bar{\eta})$ one can construct out of them:

$$\langle\phi_1\phi_2\phi_3\phi_4\rangle = f(\eta,\bar{\eta})\prod_{i<j}^{4} z_{ij}^{h/3-h_i-h_j}\,\bar{z}_{ij}^{\bar{h}/3-\bar{h}_i-\bar{h}_j}. \tag{107}$$

As we can see, the 4-point function is no longer uniquely fixed by conformal invariance.

1.2.7 Ward identities for correlation functions

Since we have symmetry and we have correlation functions of fields satisfying this symmetry, it is reasonable to ask about Ward identities of the theory. For our purposes, we will restrict ourselves to just postulating them.

For correlator $\langle X\rangle$ of n primary fields, we have two equivalent formulations of Ward identity (which we will not derive, but interested reader can find derivation in detail in [20]):

$$\begin{gathered} \delta_\epsilon\langle X\rangle = -\sum_i \left[\epsilon(w_i)\partial_{w_i} + \partial\epsilon(w_i)h_i\right]\langle X\rangle \\ \Updownarrow \quad \text{absolutely the same thing!} \\ \langle T(z)X\rangle = \sum_{i=1}^{n}\left[\frac{1}{z-w_i}\partial_{w_i} + \frac{h_i}{(z-w_i)^2}\right]\langle X\rangle + \text{reg.} \end{gathered} \tag{108}$$

where 2D momentum-energy tensor is $T(z) = -2\pi T_{zz}$, $\bar{T}(\bar{z}) = -2\pi T_{\bar{z}\bar{z}}$ and

For the global conformal transformations generated by $\{l_{\pm1}, l_0\}$, this variation must vanish, and therefore exist the following three relations on correlators of primary fields, which are global Ward identities:

$$\begin{aligned} \sum_i \partial_{w_i}\langle X\rangle &= 0 \\ \sum_i (w_i\partial_{w_i} + h_i)\langle X\rangle &= 0 \\ \sum_i (w_i^2\partial_{w_i} + 2w_ih_i)\langle X\rangle &= 0 \end{aligned} \tag{109}$$

1.2.8 Operator Product Expansion

When the position of two or more fields coincide, it is usual to have singularities in correlation functions. The operator product expansion, or OPE, is a representation of a product of operators at positions z and w, by a sum of terms, each being a [well defined as $z \to w$ operator] $\times$ [possibly divirging as $z \to w$ function of $(z-w)$].

Definition 6. *OPE of two fields is an expansion*

$$A(z)B(w) = \sum_{n=-\infty}^{N} \frac{\{AB\}_n(w)}{(z-w)^n} \tag{110}$$

as $z \to w$, where fields $\{AB\}_n(w)$ are nonsingular at $w = z$.

The most important part of the OPE is the singular part. Therefore, we will usually disregrad the regular in $(z-w)$ terms and will put the symbol when we mean equality modulo expressions regular as $z \to w$. For example, OPE of a primary field ϕ of conformal dimensions h and $\bar{h}$:

$$T(z)\phi(w,\bar{w}) \sim \frac{h}{(z-w)^2}\phi(w,\bar{w}) + \frac{1}{z-w}\partial_w\phi(w,\bar{w}), \tag{111}$$

$$\bar{T}(\bar{z})\phi(w,\bar{w}) \sim \frac{\bar{h}}{(\bar{z}-\bar{w})^2}\phi(w,\bar{w}) + \frac{1}{\bar{z}-\bar{w}}\partial_{\bar{w}}\phi(w,\bar{w}). \tag{112}$$

1.2.9 Closing remarks on CFT

Conformal field theory is a fascinating and deep area that any theoretical physicist needs to explore sooner or later. It has applications in surprisingly many domains of theoretical physics at different levels of abstraction. String theory and the AdS/CFT correspondence (as follows from the name) simply could not exist without conformal field theory.

However, in these lectures we will limit ourselves to the essentials of CFT mentioned above, hoping that they will be enough to set an introductory bridge to celestial holography.

Rigorosity Alert 2. *If between any lines that I wrote you start asking yourself the question "why?" I strongly recommend opening the Big Yellow Book [20] or a smaller blue book [21].*

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